Branch/Class/Batch: AI&DS/TE-AIDS/A1
Student Roll no.: A16
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EXPERIMENT NUMBER: 7

AIM: To configure and verify static routing and dynamic routing protocols (RIP and OSPF) in a simulated network environment using Cisco Packet Tracer.

OBJECTIVES: To establish end-to-end communication between multiple networks by configuring and validating static and dynamic routing using Cisco Packet Tracer.

THEORY:

1. What is the difference between static routing and dynamic routing?
2. What is RIP and which metric does it use to select the best route?
3. What is OSPF, and what metric does it use?
4. What is the purpose of the `ip route` command in static routing?
5. What is the purpose of the `show ip route` command, and how can you identify static, RIP, and OSPF routes?

RESOURCES REQUIRED:

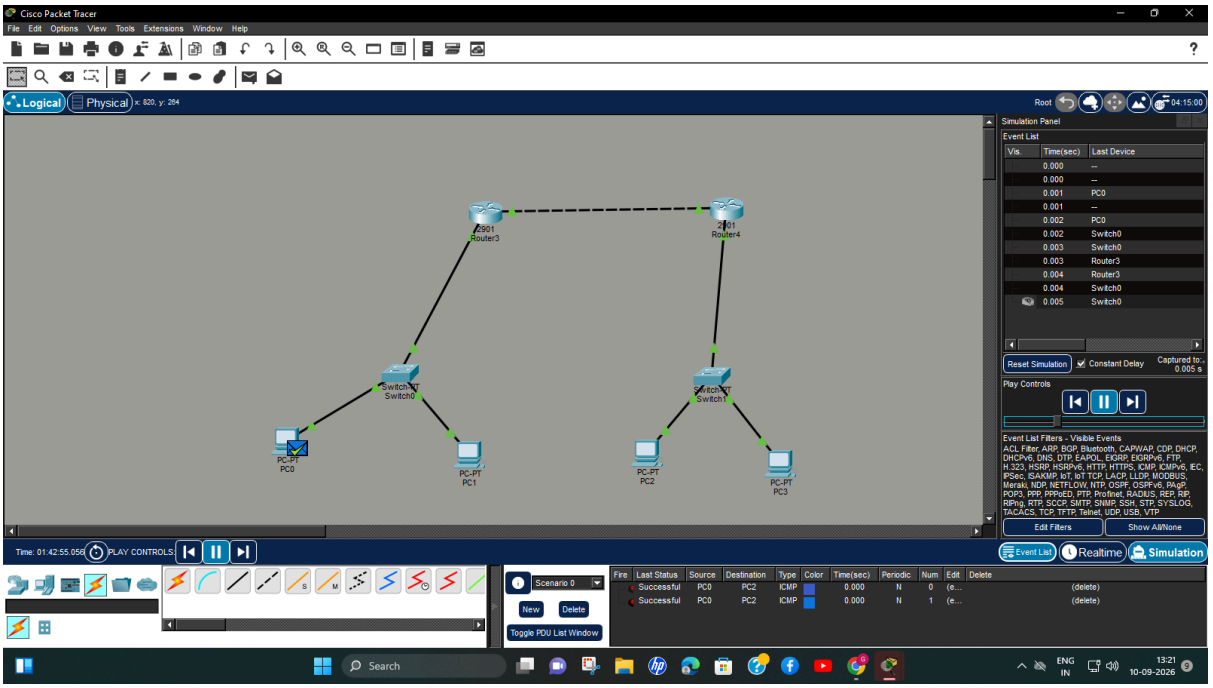
PROCEDURES:

1. Open Cisco Packet Tracer and create the required network topology using PCs, switches, and routers.
2. Connect all devices using appropriate network cables.
3. Configure the IP address, subnet mask, and default gateway for each PC.
4. Open the CLI of each router and configure the IP addresses for all connected router interfaces.
5. Enable the configured interfaces using the `no shutdown` command.
6. Verify the status and IP configuration of the router interfaces using:
   `show ip interface brief`
7. Configure static routing on the routers using the `ip route` command by specifying the destination network and next-hop address.
8. Verify the static routing entries using:
   `show ip route`
9. Test communication between PCs belonging to different networks using the `ping` command.
10. Remove the static routes and configure RIP on the routers using the appropriate network addresses.
11. Verify the RIP configuration and learned routes using:
    `show ip route`
12. Test connectivity between the different networks using the `ping` command.
13. Remove the RIP configuration and configure OSPF on the routers using the required network addresses and Area 0.
14. Verify OSPF routes using:
    `show ip route`
15. Verify OSPF neighboring routers using:
    `show ip ospf neighbor`

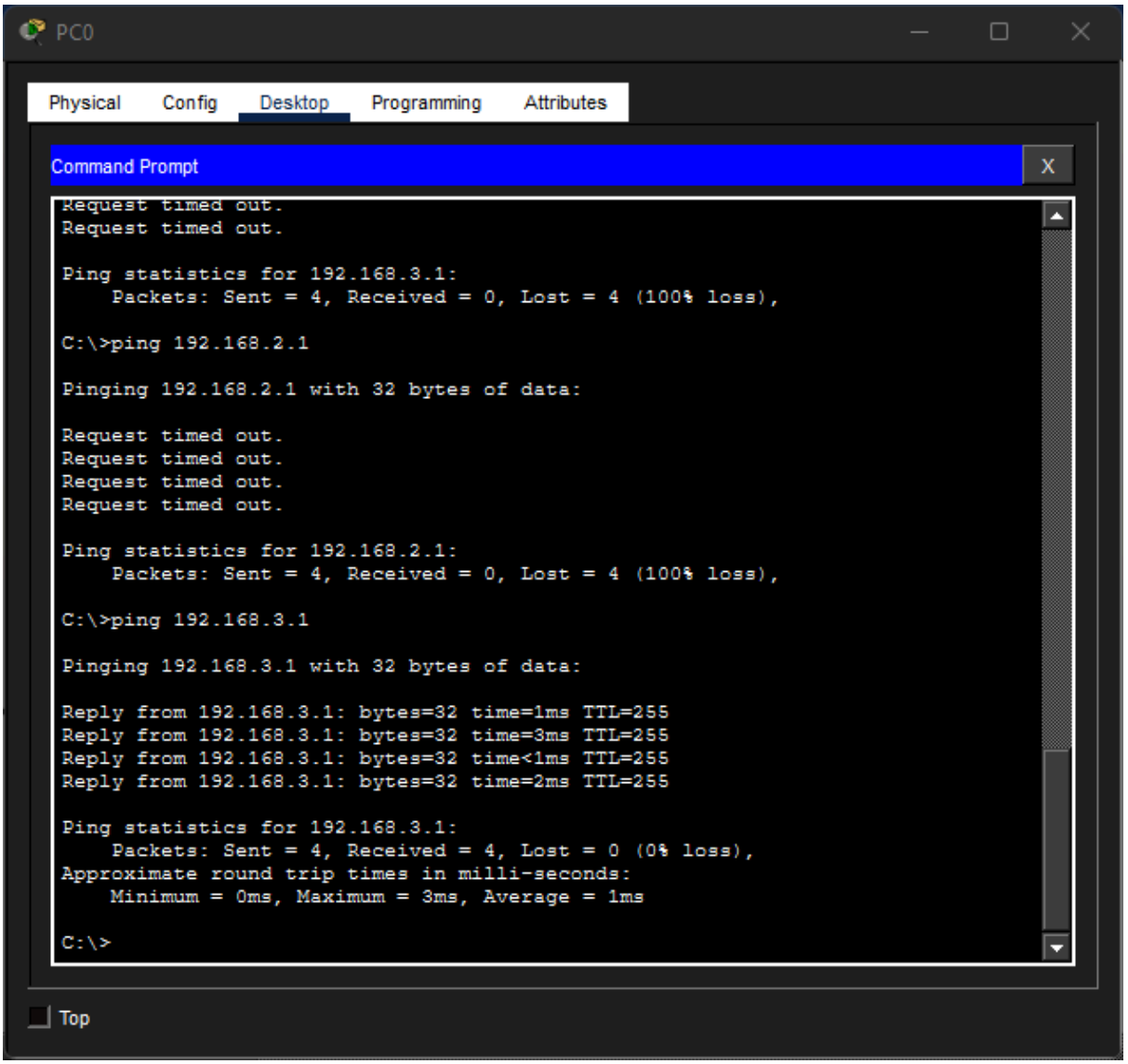
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16. Test end-to-end connectivity using `ping` and, if required, `tracert`.
17. Record the routing table entries and connectivity results for static routing, RIP, and OSPF.

RESULT:



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CONCULSION:

The experiment was successfully completed using Cisco Packet Tracer. Static routing, RIP, and OSPF were configured and verified between multiple networks. Static routing was configured manually using the **ip route** command, while RIP and OSPF dynamically learned routes from neighboring routers. The routing tables were verified using **show ip route**, and OSPF neighbor relationships were checked using **show ip ospf neighbor**. Successful ping tests confirmed end-to-end connectivity between different networks. Thus, the experiment demonstrated the configuration, verification, and practical differences between static and dynamic routing protocols.